

UBUNTU INNOVATIONS (PTY) LTD

IT Infrastructure & Support Solution

Consolidated Project Report

Prepared by: The Lab Rats

Team Members: Abethu | Nur Johnston | Hlumelo

Programme: Capaciti Tech Career Accelerator | Google IT Support Certificate

Date: June 2026 | Cape Town, South Africa

Section 1: Business Requirements Analysis

Prepared by: Abethu

1.1 Company Overview

Ubuntu Innovations (Pty) Ltd is a technology startup based in Cape Town with 25 employees distributed across multiple departments including Finance, HR, Sales, Management, and Software Development.

1.2 Organisational Structure

Department	Employees	Primary IT Needs
Executive Management	3	Email, file access, reporting tools, VPN, laptops
Finance	4	Accounting software, secure file storage, spreadsheets
Human Resources	3	HR management software, confidential file storage, email
Sales and Marketing	7	CRM, email, presentation tools, internet access
Software Development	8	IDEs, version control (Git), terminal access, local servers
Total	25	

1.3 Business Applications Required

Application	Department(s)	Purpose
Microsoft 365 / Google Workspace	All departments	Email, documents, spreadsheets, presentations
Sage Accounting / Xero	Finance	Invoicing, payroll, financial reporting
Zoho HR / BambooHR	Human Resources	Leave management, employee records, onboarding
Visual Studio Code / IntelliJ IDEA	Software Development	Code editing and development
GitHub / GitLab	Software Development	Version control, code collaboration
Slack / Microsoft Teams	All departments	Internal communication and collaboration
Veeam / Acronis	IT Administration	Backup and disaster recovery

1.4 Data Security Requirements

Ubuntu Innovations handles sensitive business, financial, and personnel data. The following data security requirements have been identified:

- All user accounts must be protected with strong passwords (minimum 12 characters, including uppercase, lowercase, numbers and special characters).

- Multi-factor authentication must be enabled for all user accounts, especially those with administrative or financial access.
- Department folders must be access controlled so that only authorised users can read, write, or modify files.
- All devices must have antivirus software installed and kept up to date.
- Sensitive data belonging to Finance and HR departments must be stored in encrypted folders.
- A regular backup schedule must be implemented to prevent data loss.

1.5 Connectivity Needs

- Wired Ethernet connections for all desktop computers and servers using Cat6 cabling for high-speed, reliable connectivity.
- Wireless Wi-Fi 6 access for laptops, mobile devices, and guest access, provided through two dual-band access points placed to cover the entire office.
- A business-grade broadband internet connection with a minimum speed of 100 Mbps to support all 25 users simultaneously.
- A VPN solution for remote access by management and authorised staff working off-site.
- The network must be segmented using VLANs to separate traffic between departments (servers, user devices, printers, and guest Wi-Fi).
- Secure remote access to cloud-hosted and local development resources.

Section 2: Hardware Inventory

Prepared by: Abethu

The following hardware has been recommended to meet the operational needs of Ubuntu Innovations. Each item is justified based on the number of employees, department requirements, and business continuity needs.

Asset	Suggested Specification	Business Justification
Desktop Computers	Intel i5, 16GB RAM	For all office-based staff in Finance, HR, Sales and Development who work at fixed workstations.
Laptops	Intel i7, 16GB RAM	For the 3 Executive Management staff and 2 senior developers who require mobility and remote access.
Router / Firewall	FortiGate 60F	Provides internet connectivity, network security, firewall protection, and VPN support for all 25 users.
Wireless Access Points	Dual-band Wi-Fi 6	Provides full office wireless coverage for laptops, mobile devices, and guest Wi-Fi. Two units ensure coverage with no dead zones.
NAS Device	2-bay or 4-bay NAS, 8TB RAID 1	Centralised file storage, daily backups, and synchronisation with offsite cloud backup storage.
Local Development Server	AMD Ryzen 7/Xeon, 32GB RAM, 2TB SSD	Hosts internal development environments, virtual machines, Git repositories, testing services, and shared developer resources.

Total Hardware Budget Estimate

Asset	Price	Quantity	Total
Desktop Computers	R 8,500	20	R 170,000
Laptops	R 16,000	5	R 80,000
Router / Firewall	R 12,000	1	R 12,000
NAS Device	R 9,500	1	R 9,500
Wireless Access Points	R 5,000	2	R 6,500
Local Development Server	R 25,000	1	R 25,000
Estimated Total			R 303,000

Section 3: Software Inventory & Licensing

Prepared by: Nur Johnston

Software	Department	License Type	Users	Annual Cost (ZAR)
Microsoft 365 / Google Workspace	All	Subscription (annual)	25	R 45,000
Antivirus (e.g. Bitdefender)	All	Subscription (annual)	25	R 15,000
Slack / Microsoft Teams	All	Subscription (annual)	25	R 30,000
Sage Accounting / Xero	Finance	Subscription (annual)	4	R 10,000
Zoho HR / BambooHR	Human Resources	Subscription (annual)	3	R 9,000
CRM (e.g. HubSpot)	Sales & Marketing	Subscription (annual)	7	R 10,500
Visual Studio Code / IntelliJ	Software Development	Free	8	R 0
Git / GitHub	Software Development	Free	8	R 0
Veeam / Acronis	IT Administration	Subscription (annual)	1	R 5,000
Total Annual Cost				R 124,500

Section 4: Network Diagram & IP Addressing

Prepared by: Nur Johnston

4.1 Network Topology Overview

Ubuntu Innovations uses a star topology with a central managed switch. The network is segmented into VLANs for security and performance isolation.

Network topology: Internet > Router/Firewall (FortiGate 60F) > Managed Switch (Cisco SG350-28) > access layer devices including desktop PCs, laptops, NAS, local development server, wireless access points (office and guest), and printers.

4.2 VLAN Configuration

VLAN ID	VLAN Name	Subnet	Purpose
1 (Default)	Internal	192.168.1.0/24	All office devices (desktops, laptops, servers, NAS, printers)
2	Guest	192.168.2.0/24	Guest Wi-Fi — completely isolated from internal network
10	Servers	192.168.10.0/24	NAS (192.168.10.10), future servers
20	User Devices	192.168.20.0/24	Desktops (.10–.29), Laptops (.30–.34)
30	Printers	192.168.30.0/24	Printer 1 (.10), Printer 2 (.11)
40	Wi-Fi	192.168.40.0/24	AP1 (.2), AP2 (.3)

4.3 Static IP Assignments

Device	Hostname	IP Address	Purpose
Router / Firewall	ubuntu-router	192.168.1.1	Default gateway, DHCP server, VPN
Network Switch	ubuntu-switch	192.168.1.2	Management access
NAS Device	ubuntu-nas	192.168.1.10	File storage and backups
Local Server	ubuntu-server	192.168.1.11	Development, database, test environment
WAP 1 (Office)	ubuntu-wap-office	192.168.1.20	Office Wi-Fi for laptops
WAP 2 (Guest)	ubuntu-wap-guest	192.168.1.21	Guest Wi-Fi (isolated)
Printer 1	ubuntu-printer-1	192.168.30.10	Shared printing (Finance wing)
Printer 2	ubuntu-printer-2	192.168.30.11	Shared printing (Dev wing)

4.4 Dynamic IP Ranges

Device Type	IP Range	Subnet	Notes
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Desktop PCs (20)	192.168.1.50 – 192.168.1.200	Internal	Wired connection
Laptops (5)	192.168.1.50 – 192.168.1.200	Internal	Wi-Fi connection
Future devices	192.168.1.201 – 192.168.1.250	Internal	For expansion
Guest devices	192.168.2.50 – 192.168.2.200	Guest	Isolated from internal network

Section 5: IT Support Troubleshooting Log — Week 1

Prepared by: Hlumelo | May 2026

5.1 Incident Summary

Ticket	Department	Issue	Resolution Time	Status
INC-001	Finance	Desktop not powering on	~25 minutes	Resolved
INC-002	Sales & Marketing	No internet connectivity	~40 minutes	Resolved
INC-003	Human Resources	Printer offline	~30 minutes	Resolved

5.2 INC-001 — Desktop Not Powering On

Field	Detail
Date Reported	19 May 2026
Department	Finance (Workstation: 192.168.20.12)
Priority	High
Resolution Time	~25 minutes

Problem Description

A Finance department employee reported that their desktop computer (Intel Core i5 12th Gen, 16 GB RAM, 512 GB SSD, Windows 11 Pro) would not turn on when the power button was pressed. The screen remained black with no fan noise, LED indicators, or POST beeps.

Root Cause & Resolution

The power cable had worked itself partially free at the PSU inlet, likely due to movement of the workstation during the recent office setup. The cable was reseated and the workstation powered on normally. A cable tie was applied to secure the cable to the desk leg to prevent recurrence. No hardware replacement was required and no data loss occurred.

Preventive Measures

- All desktop power cables to be checked and secured during the initial office setup phase.
- UPS units (APC 1500VA) confirmed operational — continue scheduled monthly UPS battery tests.

5.3 INC-002 — No Internet Connectivity

Field	Detail
Date Reported	20 May 2026
Department	Sales & Marketing (Multiple users, VLAN 20)
Priority	High
Resolution Time	~40 minutes (including ISP contact)

Problem Description

Multiple users across the Sales and Marketing department reported they could not access the internet. Internal resources including the Synology NAS and network printers remained accessible, indicating the internal LAN was functioning correctly. The outage affected HubSpot CRM, Microsoft 365, Canva Pro, and general web browsing.

Root Cause & Resolution

ISP line instability caused the ONT to lose sync, dropping the WAN connection on the FortiGate router. The ISP ONT and FortiGate 60F were restarted. WAN connectivity was restored within 40 minutes. DNS settings on the FortiGate were verified (Primary: 8.8.8.8, Secondary: 8.8.4.4).

Preventive Measures

- Evaluate adding a secondary LTE/4G failover connection to the FortiGate for business continuity during future ISP outages.
- Schedule monthly router firmware checks via the FortiGate admin panel.
- Request a formal ISP SLA document guaranteeing minimum 99.9% uptime.

5.4 INC-003 — Network Printer Offline

Field	Detail
Date Reported	21 May 2026
Department	Human Resources
Priority	Medium
Resolution Time	~30 minutes

Problem Description

An HR department employee reported that the shared network printer (HP LaserJet Pro M404dn, Printer 1, VLAN 30, static IP: 192.168.30.10) was showing as Offline in Windows 11 Devices and Printers. Print jobs submitted by HR staff were queuing but not printing.

Root Cause & Resolution

The printer's IP address had been reassigned by DHCP from 192.168.30.10 to 192.168.30.15 following a lease renewal. A permanent static IP (192.168.30.10) was assigned to the printer via its embedded web server. The printer port was updated on all HR workstations, stuck print queues were cleared, and the Print Spooler service was restarted.

Preventive Measures

- All network devices (NAS, printers, access points, switch) to be configured with static IPs during initial setup.
- DHCP scope on the FortiGate to be reviewed: exclude static IP ranges (192.168.10.x, 192.168.30.x) from the DHCP pool to prevent conflicts.

Section 6: Shared Folder Structure

Prepared by: Hlumelo | June 2026 | Document Code: HLUSTA8

The company file server is named FS01. All shared folders live under: \\FS01\\CompanyShare\\

6.1 Folder Hierarchy

The following top-level folders are established on the file server:

- \\FS01\\CompanyShare\\Executive\\ — BoardReports, StrategicPlans, Contracts
- \\FS01\\CompanyShare\\Finance\\ — Budgets, Invoices, Payroll, Reports
- \\FS01\\CompanyShare\\HumanResources\\ — EmployeeRecords, Recruitment, LeaveManagement, HRPolicies
- \\FS01\\CompanyShare\\SalesMarketing\\ — Campaigns, ClientData, SalesReports, MarketingAssets
- \\FS01\\CompanyShare\\SoftwareDevelopment\\ — Projects, Documentation, Testing, Resources
- \\FS01\\CompanyShare\\Shared\\ — Policies, Templates, Announcements
- \\FS01\\CompanyShare\\IT\\ — Documentation, Software, Backups

6.2 Permissions Summary

Folder	Exec	Finance	HR	Sales	Dev	IT
Executive	R/W	-	-	-	-	-
Finance	R	R/W	-	-	-	-
HumanResources	R	-	R/W	-	-	-
SalesMarketing	R	-	-	R/W	-	-
SoftwareDevelopment	R	-	-	-	R/W	-
Shared	R/W	R/W	R/W	R/W	R/W	R/W
IT	R	-	-	-	-	R/W

R = Read only | R/W = Read and Write | — = No access

6.3 Naming Conventions

- No spaces in file or folder names — use PascalCase (e.g. SalesReports not Sales Reports).
- No special characters such as / \ : * ? " < > |
- Add version numbers where needed (e.g. Budget_v2.xlsx).
- Use YYYY-MM-DD format for dated files (e.g. Report_2026-06-01.pdf).

Section 7: User & Permissions Matrix

Prepared by: Abethu

The permissions matrix controls which department groups can access specific company folders. Management has full access to all folders, while other departments only access folders relevant to their responsibilities.

Resource	mgmt_grp	finance_grp	hr_grp	sales_grp	dev_grp
/Company/Public	X	X	X	X	X
/Company/Finance	X	X	-	-	-
/Company/HR	X	-	X	-	-
/Company/Sales	X	-	-	X	-
/Company/Development	X	-	-	-	X
/Company/Management	X	-	-	-	-

Section 8: Software Deployment Plan

Prepared by: Nur Johnston | The Lab Rats

This document outlines the steps for deployment, updating, and management of software for all 25 machines at Ubuntu Innovations. The goal is to maintain consistency, security, and legally compliant software usage across the organisation.

8.1 Standard Software (All Users)

Software	Purpose	Deployment Method	Update Frequency
Microsoft 365 Business Premium	Email, Word, Excel, PowerPoint	Cloud / Intune	Automatic (monthly)
Bitdefender Antivirus	Security and malware protection	Cloud push	Daily signature updates
Slack Pro	Internal communication	Intune / Direct download	Automatic
Zoom Business	Video conferencing	Intune / Direct download	Automatic
Google Chrome	Web browsing	Intune / Group Policy	Automatic
Mozilla Firefox	Web browsing (alternative)	Intune / Group Policy	Automatic

8.2 Department-Specific Software

Department	Software	Licenses	Deployment
Finance	Sage Business Cloud Accounting	4	Cloud (vendor managed)
Human Resources	BambooHR	3	Cloud (vendor managed)
Sales & Marketing	HubSpot Sales Hub	7	Cloud (vendor managed)
Executive Management	Microsoft Project Online	3	Intune
Finance, Exec, HR	Adobe Acrobat Pro	10	Intune

8.3 Development Team Software

Software	Purpose	License	Responsible
Ubuntu Server 22.04 LTS	Operating system	Open source	IT Support
MySQL / PostgreSQL	Database for development	Open source	IT Support + Dev Lead

GitLab CE	Local code repository mirror	Open source	IT Support
Docker Engine	Container hosting	Free	IT Support + Dev Lead
Visual Studio Code	Code editing	Free	Developer (self-install)
Git	Version control	Free	Developer (self-install)
Docker Desktop	Container management	Free	Developer (self-install)
Node.js / Python	Runtime environments	Free	Developer (self-install)
Postman	API testing	Free	Developer (self-install)

8.4 Update Procedures

Software Category	Update Schedule	Responsible Party
Windows 11 OS	Monthly (Patch Tuesday)	IT Support
Microsoft 365	Automatic (monthly)	Microsoft / IT Support
Bitdefender Antivirus	Daily (automatic)	Automatic
Department SaaS (Sage, BambooHR, HubSpot)	Vendor-managed (automatic)	Vendor
Development tools	Quarterly manual review	Development Team Lead
Critical security patches	Within 48 hours of release	IT Support
Local server (Ubuntu)	Weekly (Sundays at 2am)	IT Support (automated)

8.5 Total Annual Software Cost

Category	Annual Cost (ZAR)
Standard software (25 users)	R 102,000
Department-specific software	R 29,500
Development tools (free / open source)	R 0
Total	R 131,500

Section 9: Command-Line Reference Guide

Prepared by: Hlumelo | June 2026 | Document Code: HLUSTA9

This guide is a reference for the IT support team at Ubuntu Innovations. It covers the most common command-line tasks for managing Ubuntu Linux systems and Windows workstations — including user management, permissions, network checks, and system monitoring.

9.1 Ubuntu Linux — Quick Reference

Task	Linux Command
Create a new user	<code>sudo adduser jdoe</code>
Change a user's password	<code>sudo passwd jdoe</code>
Add user to a group	<code>sudo usermod -aG finance jdoe</code>
Check user's groups	<code>groups jdoe</code>
Delete user and home folder	<code>sudo deluser --remove-home jdoe</code>
List all users	<code>cut -d: -f1 /etc/passwd</code>
Set file permissions	<code>chmod 640 report.pdf</code>
Set folder permissions	<code>sudo chmod -R 750 /mnt/companyshare/Finance</code>
Change file ownership	<code>sudo chown jdoe:finance report.pdf</code>
View IP address	<code>ip addr show</code>
Ping a host	<code>ping 8.8.8.8</code>
Trace route	<code>tracert google.com</code>
View open ports	<code>ss -tuln</code>
Check disk space	<code>df -h</code>
Check RAM usage	<code>free -h</code>
Restart a service	<code>sudo systemctl restart networking</code>
View system logs	<code>sudo tail -f /var/log/syslog</code>
Update packages	<code>sudo apt update && sudo apt upgrade</code>
Install a package	<code>sudo apt install openssh-server</code>
Map network drive	<code>net use Z: \\FS01\CompanyShare /persistent:yes</code>

9.2 Windows CMD — Quick Reference

Task	Windows Command
Create a local user	<code>net user jdoe P@ssw0rd123 /add</code>
Add user to group	<code>net localgroup Finance jdoe /add</code>
Delete a user	<code>net user jdoe /delete</code>

List all users	net user
Check IP config	ipconfig /all
Ping a host	ping 8.8.8.8
Trace route	tracert google.com
Flush DNS cache	ipconfig /flushdns
View active connections	netstat -an
List running processes	tasklist
Kill a process	taskkill /IM chrome.exe /F
Check disk space	wmic logicaldisk get size,freespace,caption
System info	systeminfo
Restart PC	shutdown /r /t 0

9.3 Security Reminders for IT Support

- Only use sudo when you actually need it — do not run everything as root.
- Do not type passwords directly into commands where they will appear in command history.
- Remove user accounts as soon as someone leaves the company.
- Give users the minimum permissions they need to do their job (principle of least privilege).
- Keep a log of any admin-level commands run on production systems.

Section 10: Cybersecurity Policy

Prepared by: Abethu | Week 3

The purpose of this Cybersecurity Policy is to protect the organisation's information systems, networks, devices, and data from unauthorised access, cyber threats, and security breaches.

10.1 Password Complexity Requirements

- Minimum of 12 characters.
- Include uppercase and lowercase letters.
- Include numbers and special characters.
- Passwords must be changed every 90 days.
- Password sharing is strictly prohibited.

10.2 Multi-Factor Authentication (MFA)

MFA must be enabled for: email accounts, cloud services, administrative accounts, and remote access systems.

Users must verify their identity using one of: Password + Mobile Authentication App, or Password + One-Time PIN (OTP).

10.3 Device Encryption

- Full disk encryption must be enabled on laptops and desktops.
- Mobile devices must use encryption and screen locks.
- USB drives containing company data must be encrypted.
- Lost or stolen devices must be reported immediately.

10.4 Antivirus Standards

- Approved antivirus software must be installed on all devices.
- Real-time protection must remain enabled.
- Virus definitions must update automatically.
- Weekly system scans must be performed.
- Any detected threats must be reported to IT immediately.

10.5 Patch Management

- Operating systems must be updated regularly.
- Security patches must be installed within 7 days of release.
- Critical security updates must be applied immediately.
- Software inventories must be maintained.
- Unsupported software must be removed or upgraded.

10.6 Acceptable Use Policy

Acceptable activities: business-related communication, accessing work-related resources, using approved software and applications.

Prohibited activities: downloading unauthorised software, accessing malicious or illegal websites, sharing confidential company information, disabling security controls, using company systems for illegal activities.

Violations may result in disciplinary action.

10.7 Risk Assessment Matrix

Risk	Likelihood	Impact	Mitigation
Phishing Attacks	High	High	Security awareness training
Ransomware	Medium	High	Backups and EDR (Endpoint Detection & Response)
Hardware Failure	Medium	Medium	Redundancy and regular maintenance
Power Outage	Medium	High	UPS units on all critical systems

Section 11: Incident Response Plan

Prepared by: Hlumelo | June 2026

This is the incident response plan for Ubuntu Innovations. Its purpose is to ensure that if something goes wrong — a cyberattack, data breach, malware, or any other incident — there is a clear process to follow for a fast and organised response.

11.1 Scope

This plan covers all IT systems at Ubuntu Innovations, including: workstations and laptops, the file server (FS01), network infrastructure (router, switches, firewall), cloud services and company email accounts, and any personal devices used to access company systems.

11.2 Incident Response Team

Role	Responsibility
IT Support Lead	Overall coordination of the response
IT Support Technician	Technical investigation and containment
Department Manager	Communication with affected department
Executive Manager	Final decisions on escalation or legal action
HR Representative	Handles any incidents involving staff conduct

11.3 Incident Severity Levels

Level	Description	Example
Low	Minor issue, no data at risk	Failed login attempt from outside
Medium	Possible data exposure or system compromise	Malware detected on one workstation
High	Confirmed breach or major disruption	Ransomware spreading across network
Critical	Full system compromise or data leak confirmed	Database exfiltrated, multiple systems down

11.4 The 6 Phases of Response

Phase 1 – Preparation

Keep this document up to date, ensure all staff know how to report an incident, keep system and access logs turned on, maintain offline backups tested regularly, and run security awareness training at least once a year.

Phase 2 – Identification

Determine who noticed the incident and when, what systems are involved, whether it is still ongoing, and how serious it appears. Report immediately to the IT Support Lead — do not attempt to fix it yourself.

Phase 3 – Containment

Short-term: disconnect affected device(s) from the network, change passwords for affected accounts, block suspicious IP addresses at the firewall. Long-term: set up a clean temporary system if needed, increase monitoring on similar systems.

Phase 4 – Eradication

Remove malware, delete files or accounts created by the attacker, patch exploited vulnerabilities, reset credentials for all potentially affected accounts, and review logs to confirm no backdoors remain.

Phase 5 – Recovery

Restore systems from clean backups, reconnect systems to the network one at a time and monitor closely, confirm everything is working before declaring the incident closed, and maintain extra monitoring for at least 2 weeks after recovery.

Phase 6 – Lessons Learned

Within 2 weeks of closing an incident, hold a short review meeting. Document what happened, what worked well, what could have been done faster, and what changes need to be made. Add a written summary to the incident log.

Section 12: Backup & Disaster Recovery Plan

Prepared by: Nur Johnston | The Lab Rats

This document outlines all steps for the backup and disaster recovery process at Ubuntu Innovations. The goal is to ensure the business keeps running in cases of hardware failure, load shedding, data loss, cyberattack, or natural disaster.

12.1 What Gets Backed Up

Device	IP Address	Data to Back Up	Frequency
NAS Device	192.168.1.10	All shared folders (/Finance, /HR, /Sales, /Development, /Management)	Daily incremental, Weekly full
Local Server	192.168.1.11	Databases (MySQL/PostgreSQL), GitLab repositories, Docker volumes	Daily incremental
Desktop PCs (20)	192.168.1.50–200	User profiles (Documents, Desktop, Downloads)	Weekly (user-initiated)
Laptops (5)	192.168.1.50–200	User profiles (Documents, Desktop, Downloads)	Weekly (user-initiated)

12.2 Backup Schedule

Backup Type	Frequency	Time	Storage Location	Retention Period
Incremental	Daily (Mon–Fri)	2:00 AM	Local NAS (RAID 1)	30 days
Full	Weekly	Sunday at 1:00 AM	Local NAS + Cloud	4 weeks (local), 12 months (cloud)
Monthly Archive	Monthly	1st Sunday at 2:00 AM	Cloud only (AWS S3 / Backblaze)	12 months

12.3 Storage Locations

Location	Device	Capacity	Purpose
Primary (Local)	Synology NAS (8TB RAID 1)	4TB usable (mirrored)	Daily and weekly backups
Secondary (Local)	Local server (2x1TB SSD RAID 1)	1TB usable	Critical database and GitLab backups only
Tertiary (Cloud)	AWS S3 or Backblaze B2	2TB	Offsite monthly archives and disaster recovery

12.4 Recovery Objectives

Objective	Target	Explanation
RTO (Recovery Time Objective)	4 hours	Maximum time allowed to restore critical systems after a disaster
RPO (Recovery Point Objective)	1 hour	Maximum acceptable data loss — no more than 1 hour of work

12.5 Recovery Procedures

a. Single file accidentally deleted

User contacts IT Support → IT Support opens NAS backup interface → Locate file from most recent backup → Restore to original location → Confirm with user.

b. NAS device failure

IT Support detects NAS failure via monitoring alert → Replace failed hard drive (RAID 1 allows hot swap) → NAS automatically rebuilds data → If both drives fail, restore from cloud backup → Verify shared folders are accessible to all departments.

c. Digital attack (ransomware / files encrypted)

Identify infected device(s) and disconnect from network → Identify last known clean backup → Wipe infected devices completely → Restore all data from clean backup (cloud or NAS) → Scan all restored files for malware → Reconnect devices and reset user passwords.

d. Load shedding

UPS (1500VA) keeps router, switch, NAS, and local server running → UPS provides 15–30 minutes of runtime → If power not restored within 15 minutes, IT Support initiates graceful shutdown of NAS and server → When power returns, systems auto-restart.

e. Fire, flood, or lockdown

Employees work from home (remote) → Access cloud backups (AWS S3 / Backblaze) from any location → Set up temporary VPN access to cloud-hosted critical data → Restore essential files to employee laptops → When office is accessible, rebuild local infrastructure and restore from cloud.

12.6 Backup Testing Schedule

Test Type	Frequency	Procedure	Responsible
Automated integrity check	Daily	Backup software verifies backup files are readable	Automatic
Test file restore	Weekly	Restore one random file from backup to verify	IT Support
Full disaster recovery drill	Quarterly (Jan, Apr, Jul, Oct)	Simulate NAS failure and restore from cloud	IT Support + Management

Section 13: Security Awareness Training Outline

Prepared by: Hlumelo | June 2026

This document is an outline for the security awareness training that all Ubuntu Innovations staff need to complete. The goal is to ensure everyone in the company knows enough about cybersecurity to prevent the most common types of security incidents — most breaches happen because someone clicked a bad link, used a weak password, or plugged in an unknown USB drive.

13.1 Training Overview

Detail	Info
Target Audience	All staff (all departments) — 25 staff members
Frequency	Once a year minimum, with short refreshers every 6 months
Format	In-person session + short online quiz
Duration	About 2 hours for the full session
Facilitator	IT Support Lead
Completion Tracking	Signed attendance sheet + quiz results saved to HR records

13.2 Training Modules

Module	Topic	Duration
Module 1	Why Cybersecurity Matters to Us	~15 min
Module 2	Passwords and Account Security	~20 min
Module 3	Phishing and Social Engineering	~30 min
Module 4	Safe Internet and Email Use	~15 min
Module 5	Physical Security	~10 min
Module 6	Incident Reporting	~15 min
Module 7	Remote Work Security	~15 min

13.3 Key Module Highlights

Module 3 – Phishing and Social Engineering (most critical)

How to spot a phishing email: check that the sender address matches the display name, be wary of urgent language, hover over links before clicking, do not open attachments from unknown senders, and never provide login credentials or personal information via email.

If a suspicious email is received: do not click any links, do not reply, forward to IT immediately. If a link was already clicked, report it straight away — no judgement.

Module 2 – Passwords and Account Security

Company password policy: minimum 12 characters, mix of uppercase/lowercase/numbers/symbols, no password reuse between accounts, no sharing passwords with colleagues. Staff are encouraged to use a password manager and must enable MFA on all company accounts.

13.4 Assessment

After training, all staff complete a short quiz (10–15 questions). Pass mark: 80%. Staff who do not pass receive a one-on-one follow-up session with IT. Results are kept on file in HR.

13.5 Training Schedule

Session	Audience	When
Annual Full Training	All staff	January each year
New Staff Induction	New joiners	Within first week of joining
6-Month Refresher	All staff	July each year
Post-Incident Training	Relevant staff	After any Medium+ security incident

Related Documents

The following documents are referenced across the tasks in this report:

- Task 4 – Network Diagram
- HLUSTA7 – User and Permissions Matrix
- HLUSTA8 – Shared Folder Structure
- HLUSTA9 – Command-Line Guide
- Task 13 – Incident Response Plan
- Task 14 – Disaster Recovery Plan
- Task 15 – Security Awareness Training Outline